

Question	Answer	Marks
1(a)	Apply Principle of conservation of linear momentum, Taking to the right as positive, $2.0 \cdot 10 + 3.0 \cdot 5.0 = (2.0 + 3.0)v$ $v = 7.0 \text{ m s}^{-1}$	[1] [1]
1(b)	By conservation of energy, Method 1: Loss in kinetic energy of system = Gain in elastic potential energy of spring $\frac{1}{2}m_A v_i^2 + \frac{1}{2}m_B v_i^2 - \frac{1}{2}m_{A+B} v^2 = \frac{1}{2}kx^2$ $\frac{1}{2}(2.0)(10)^2 + \frac{1}{2}(3.0)(5.0)^2 - \frac{1}{2}(2.0 + 3.0)(7.0)^2 = \frac{1}{2}(1000)x^2$ $x = 0.173 \text{ m} = 0.17 \text{ m}$ Method 2: Initial total energy = Final total energy $\frac{1}{2}(2.0)(10)^2 + \frac{1}{2}(3.0)(5.0)^2 = \frac{1}{2}kx^2 + \frac{1}{2}(2.0)(7.0)^2 + \frac{1}{2}(3.0)(7.0)^2$ $x = 0.173 \text{ m} = 0.17 \text{ m}$	[1] [1]
1(c)	Apply Principle of conservation of linear momentum, Taking to the right as positive, $2.0 \cdot 10 + 3.0 \cdot 5.0 = (2.0 \cdot v_A) + (3.0 \cdot v_B)$ $2 v_A + 3 v_B = 35 \quad \text{-----}(1)$ As collision is elastic, relative speed of approach = relative speed of separation, $10 - 5.0 = v_B - v_A \quad \text{-----}(2)$ Solving (1) & (2), $v_A = 4.0 \text{ m s}^{-1}$ and $v_B = 9.0 \text{ m s}^{-1}$	[1] [1] [1]
	Max Marks	7

2020

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

Question	Answer	Marks
2(a)	● Weight is pointing downwards from centre of rod	[1]